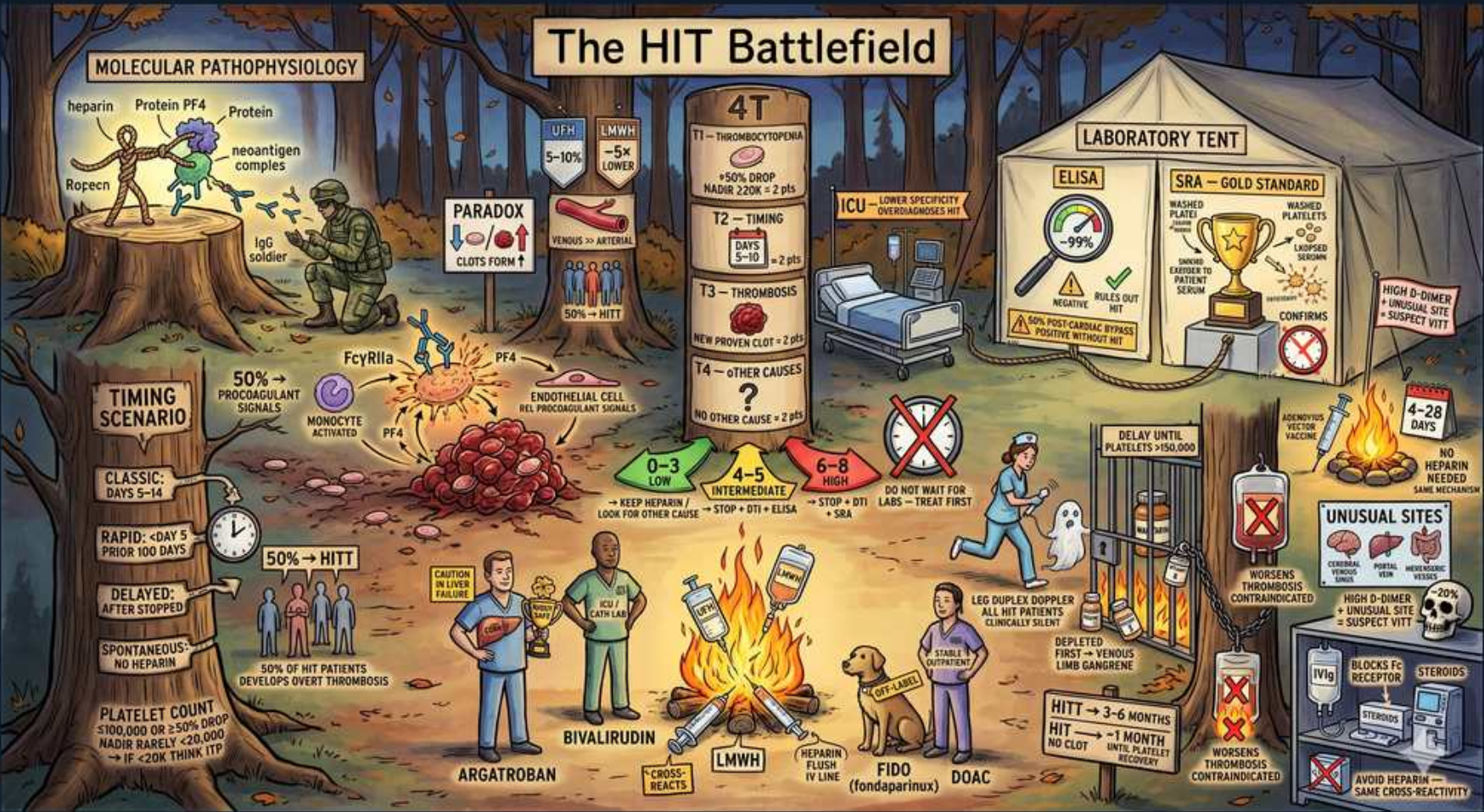


Platelet disorders — HIT (The HIT Battlefield)



1. Heparin-PF4 complex

WHAT YOU SEE

Heparin, PF4, and an antibody clipping together into one 'NEOANTIGEN COMPLEX' on the tree stump — heparin binds PF4 to form the immunogenic complex that IgG attacks.

WHAT IT ENCODES

HIT type II is an immune reaction: heparin binds platelet factor 4 (PF4) released from platelet alpha-granules, forming a heparin–PF4 complex that becomes immunogenic. IgG antibodies form against this complex (typically by days 5–10 of first exposure). Unfractionated heparin carries the highest risk (>LMWH > fondaparinux ≈ none) because larger, more negatively charged molecules form bigger PF4 complexes. This contrasts with benign HIT type I (mild, early, non-immune, self-resolving).

BOARD TRAP

Don't confuse HIT type I with type II — type I is a transient, non-immune, mild count dip in the first 1–2 days that resolves despite continued heparin and needs no action. Stopping heparin and starting a DTI for a type I dip is over-treatment; the dangerous, antibody-mediated entity is type II.

2. Prothrombotic paradox

WHAT YOU SEE

A PARADOX panel showing clots forming even as platelets fall ('CLOTS FORM!') — the hallmark twist of HIT: thrombosis despite a dropping count.

WHAT IT ENCODES

The defining paradox of HIT: despite a FALLING platelet count, the patient is hypercoagulable and forms clots (the antibodies activate platelets and generate thrombin). Thrombosis occurs in up to 30–50% of HIT — venous (DVT/PE, most common) more than arterial (limb ischemia, stroke, MI) — and is the main cause of morbidity/mortality. The thrombocytopenia is rarely severe enough to cause spontaneous bleeding.

BOARD TRAP

Don't treat low platelets in HIT as a bleeding problem — risk is clotting. The classic wrong answer is transfusing platelets (adds fuel for thrombosis) or simply holding anticoagulation; the correct move is to STOP heparin and START a non-heparin anticoagulant because the patient is actively prothrombotic.

3. Platelet/monocyte activation

WHAT YOU SEE

A blazing sunburst of platelets and monocytes firing via FcγR1a receptors — the immune complexes cross-link FcγR1a, igniting platelet activation and a thrombin burst.

WHAT IT ENCODES

The IgG–heparin–PF4 immune complexes cross-link platelet FcγR1a receptors, causing massive platelet activation, aggregation (consumption → thrombocytopenia), and release of procoagulant microparticles. The complexes also activate monocytes and endothelium to express tissue factor, driving a thrombin burst. This explains the simultaneous platelet drop AND thrombosis — platelets are consumed into clots, not destroyed for clearance alone.

BOARD TRAP

Don't expect the count to be the danger — the activated procoagulant state (thrombin generation) is. A 'reassuring' platelet count of 80–100K in a heparin patient with a new clot does not exclude HIT; the thrombosis is the alarm even when thrombocytopenia is modest.

4. 4T score

WHAT YOU SEE

A tall '4T' scoring column listing Thrombocytopenia / Timing / Thrombosis / oTher, with an ICU bed — the pretest-probability score that gates HIT testing.

WHAT IT ENCODES

The 4Ts score estimates pretest probability (0–2 points each, max 8): Thrombocytopenia (>50% fall and nadir ≥20K = 2), Timing (onset day 5–10 or ≤1 day with recent heparin = 2), Thrombosis (new confirmed thrombosis/skin necrosis = 2), and oTher cause absent (= 2). Score 0–3 = low (HIT effectively excluded, ~<5%; no need to test); 4–5 = intermediate; 6–8 = high. Intermediate/high → stop heparin, start non-heparin anticoagulant, and send immunoassay.

BOARD TRAP

Don't order a PF4 ELISA in a low-probability (4Ts 0–3) patient — its low specificity yields false positives that trigger unnecessary, costly, bleeding-risk anticoagulation. Use the 4Ts to gate testing; a low score makes HIT unlikely without any assay.

5. 50% platelet drop

WHAT YOU SEE

A timeline panel marking a '~50% DROP' from baseline — HIT's signature is a relative >50% fall (e.g., 400K to 150K), not a crash below 20K.

WHAT IT ENCODES

HIT characteristically produces a >50% fall from the patient's baseline platelet count, with a typical nadir of ~50–70K (median ~60K). The drop is relative — a patient going from 400K to 150K is 'normal range' but has fallen >50% and may have HIT. Counts <20K are distinctly unusual for HIT and should prompt another diagnosis (DIC, other drug, post-transfusion purpura).

BOARD TRAP

Severe <20K thrombocytopenia argues against HIT — that depth suggests another cause (drug-induced immune thrombocytopenia, DIC, ITP, post-transfusion purpura). Equally, judge the drop relative to baseline, not the absolute range: a fall from 400K→150K is a >50% HIT-range drop even though 150K 'looks normal.'

6. Timing 5-10 days

WHAT YOU SEE

A clock marked 'CLASSIC 5-10 DAYS' with a 'RAPID IF PRIOR EXPOSURE' note — onset day 5-10 on first exposure, within hours if heparin was given in the last ~100 days.

WHAT IT ENCODES

Classic HIT onset is day 5–10 after starting heparin (time to form IgG). RAPID-onset HIT occurs within hours to <1 day if the patient had heparin exposure within the prior ~100 days (pre-existing antibodies). DELAYED-onset HIT can begin days to weeks after heparin is stopped. Knowing the exposure timeline is essential to fit the picture and to assign the Timing points in the 4Ts.

BOARD TRAP

Don't exclude HIT because a drop happened on day 1 — that's expected rapid-onset HIT if there was heparin in the last 100 days. Likewise, a drop appearing after heparin was already stopped can still be delayed-onset HIT; the recent-exposure history reframes 'wrong' timing as classic.

7. ELISA screening

WHAT YOU SEE

A magnifying glass over an ELISA plate labeled high-sensitivity (~99%) — the screen that RULES OUT HIT when negative but over-calls when weakly positive.

WHAT IT ENCODES

The PF4–heparin antibody immunoassay (ELISA) is the screening test: HIGH sensitivity but LOW specificity, so a negative result effectively excludes HIT but a positive must be interpreted with the optical density (OD) — higher OD = more likely true/functional HIT. Many patients (especially post-cardiac-surgery) have non-pathogenic antibodies, producing false positives. A positive ELISA with intermediate/high 4Ts should be confirmed with a functional assay.

BOARD TRAP

Don't accept a weakly positive ELISA as proof of HIT — its poor specificity over-diagnoses, especially after cardiac surgery. Use it to RULE OUT (negative excludes) and confirm true positives with the functional serotonin release assay; treating on a low-OD ELISA alone leads to unnecessary anticoagulation.

8. SRA gold standard

WHAT YOU SEE

A gold trophy beside 'SRA — GOLD STANDARD' — the functional serotonin-release assay that confirms HIT with both high sensitivity and specificity.

WHAT IT ENCODES

The serotonin release assay (SRA) — a functional test showing heparin-dependent platelet activation (radiolabeled serotonin release) at therapeutic but not high heparin concentrations — is the confirmatory GOLD STANDARD with high sensitivity AND specificity. Because it is technically demanding and not rapid, it confirms the diagnosis but does NOT delay treatment: empiric management starts on clinical/4Ts grounds while the SRA is pending.

BOARD TRAP

Don't wait for the SRA before acting — it takes days and is sent out. If the 4Ts are intermediate/high, stop heparin and start a non-heparin anticoagulant immediately; treating only after SRA confirmation risks thrombosis during the delay.

9. STOP heparin

WHAT YOU SEE

Heparin and LMWH vials slashed with a red X — discontinue ALL heparin sources, including flushes and LMWH.

WHAT IT ENCODES

First and immediate step in suspected HIT: STOP ALL heparin — IV/SC unfractionated heparin, LMWH, heparin line flushes, and heparin-bonded catheters. Stopping heparin alone, however, is insufficient because the prothrombotic state persists for days; a non-heparin anticoagulant must be added. Document the diagnosis prominently and label the chart to prevent future re-exposure.

BOARD TRAP

Continuing/switching to LMWH does not treat HIT — LMWH cross-reacts with HIT antibodies and perpetuates the disease. The complete first step is removing every heparin source (including flushes), not down-titrating or substituting another heparin product.

10. Argatroban

WHAT YOU SEE

An Argatroban soldier on the battlefield — the IV direct thrombin inhibitor (hepatically cleared) preferred in renal failure for acute HIT.

WHAT IT ENCODES

Argatroban is a direct thrombin inhibitor (DTI) given as a continuous IV infusion, titrated to aPTT, and is hepatically cleared — making it the preferred agent in RENAL failure. Note it artificially prolongs the INR, which complicates later warfarin overlap. It is a mainstay for acute HIT anticoagulation in hospitalized/critically ill patients who need an IV, easily titratable option.

BOARD TRAP

Don't use argatroban (hepatically cleared) in significant hepatic impairment — choose bivalirudin instead. And remember argatroban falsely elevates INR; bridging to warfarin requires a higher target INR and platelet recovery first, or you'll stop the DTI prematurely and leave the patient unprotected.

11. Bivalirudin / DOAC

WHAT YOU SEE

A row of allied soldiers — Bivalirudin, Fido the fondaparinux dog, and a DOAC — the non-heparin anticoagulant alternatives for HIT.

WHAT IT ENCODES

Alternatives for HIT anticoagulation: bivalirudin (IV DTI, enzymatically/renally cleared — favored in hepatic impairment and periprocedurally), fondaparinux (SC factor Xa inhibitor, no cross-reactivity, used off-label especially in stable patients), and DOACs (rivaroxaban/apixaban) — increasingly used for clinically stable HIT and for transition/outpatient therapy. Choice depends on organ function, acuity, and need for titratability.

BOARD TRAP

Don't give a DOAC to an unstable, critically ill HIT patient who may need procedures or has unreliable absorption — a titratable IV DTI (argatroban/bivalirudin) is safer acutely. DOACs are best reserved for clinically stable patients without thrombosis or once stabilized.

12. Avoid warfarin early

WHAT YOU SEE

A warfarin caution sign warning of 'VENOUS LIMB GANGRENE' — starting warfarin too early drops protein C first and worsens clotting.

WHAT IT ENCODES

Do NOT start warfarin in acute HIT until platelets have recovered (typically >150K) and the patient is on a therapeutic non-heparin anticoagulant. Warfarin acutely depletes protein C (short half-life) before depleting the prothrombotic factors, transiently worsening the hypercoagulable state and risking venous limb gangrene and warfarin skin necrosis. When started, overlap with the DTI for ≥5 days and until INR is therapeutic.

BOARD TRAP

Starting warfarin alone in acute HIT causes skin necrosis / limb gangrene — protein C falls first, deepening thrombosis. If warfarin was already inadvertently started, reverse it with vitamin K and cover with a non-heparin anticoagulant; never bridge HIT with warfarin monotherapy.

13. Anticoagulate even without clot

WHAT YOU SEE

A banner 'CONTINUE ANTICOAGULATION 1-3 MONTHS' — every HIT patient needs a non-heparin anticoagulant even with no clot found.

WHAT IT ENCODES

ALL HIT patients need therapeutic non-heparin anticoagulation, even with no clot detected, because thrombotic risk persists for weeks. If no thrombosis is found, screen (e.g., lower-extremity Doppler) and continue anticoagulation for ~4 weeks (until platelets recover and antibodies wane); if thrombosis IS present (HITT), treat for at least 3 months. The clinical entity, not just the lab, drives the duration.

BOARD TRAP

Just stopping heparin is insufficient — must add an alternative anticoagulant. 'Isolated HIT' (thrombocytopenia without overt clot) is NOT observed without anticoagulation; the prothrombotic window means ~half will clot if untreated. Withholding anticoagulation until a clot appears is the wrong, dangerous choice.

14. Avoid heparin re-exposure

WHAT YOU SEE

A skull warning to 'AVOID FUTURE HEPARIN' — document HIT and never re-expose, since re-challenge triggers rapid, severe HIT.

WHAT IT ENCODES

Document HIT prominently as an allergy/adverse reaction and avoid all future heparin exposure given cross-reactivity and the risk of rapid-onset HIT on re-challenge. HIT antibodies are typically transient (wane over ~3 months); in rare unavoidable situations (e.g., cardiac surgery) where antibodies have cleared and SRA is negative, a single intraoperative heparin dose may be considered by specialists — otherwise use a non-heparin alternative.

BOARD TRAP

Don't assume 'it was a long time ago, heparin is fine now' — re-exposure while antibodies persist (or anamnестically) triggers rapid, severe HIT. Lifelong avoidance and clear documentation are the safe default; any exception requires confirmed antibody clearance and specialist involvement.
